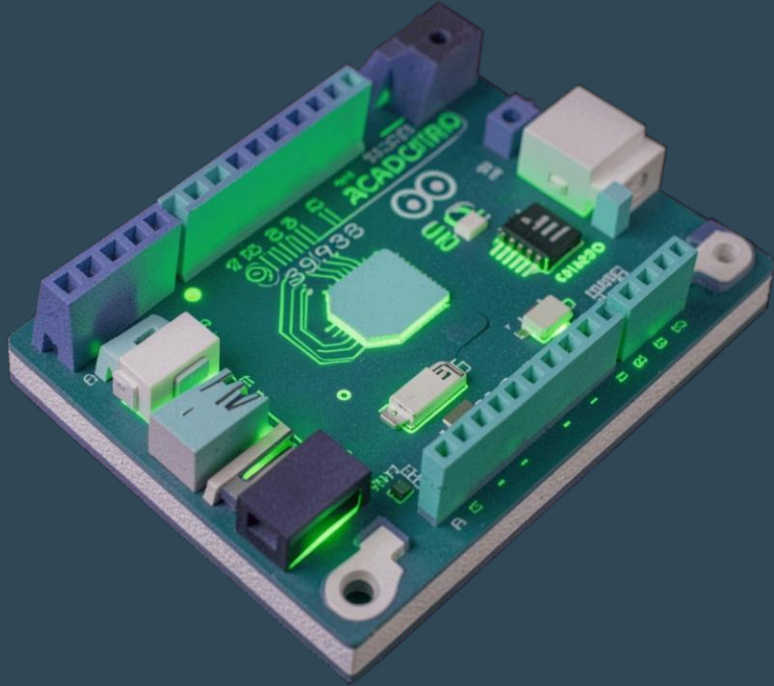


RN HIGH SCHOOL • TECH FEST 2026



Arduino UNO Based Countdown Timer

A smart countdown timer built using Arduino UNO for accurate and easy time management.



INTRODUCTION

What Is a Countdown Timer?

A countdown timer starts from a set time and counts down to zero. When the time ends, it gives an alert with a buzzer or display.

Why Is It Useful?

- Helps manage time.
- Improves focus.
- Prevents delays.

Project Objective

- Build a low-cost Arduino UNO countdown timer.
- Show the remaining time on the LCD screen.
- Use a rotary encoder to set the time.
- Sound a buzzer when the timer ends.

Key Features

- Easy to use.
- Accurate timing.
- Low cost.
- Portable.
- Useful for schools, kitchens, study sessions, exams, and workshops.



Simulation Platform

This project was designed, wired, programmed, and tested using the Wokwi Arduino Simulator. The same circuit can be built on real Arduino UNO hardware with the listed components.

PROBLEM STATEMENT

Problems Without a Countdown Timer

Without a countdown timer, managing time becomes difficult during study, exams, cooking, and daily activities.

Our Arduino UNO Countdown Timer gives accurate timing with a clear LCD display and buzzer alert.

✗ Without a Timer

- Difficulty manage time
- Miss important deadlines
- Get distracted by mobile phones
- No countdown display
- No buzzer alert
- Lower productivity

✓ With Our Timer

- Accurate countdown timer
- Clear LCD display
- Buzzer alert when time ends
- LED status indication
- Easy to set the timer
- Portable and easy to use

📌 **Real-Life Example:** During school quizzes, science practicals, or exams, students can set the timer and complete their activity on time without using a mobile phone.

COMPONENTS USED

Hardware Components Used (Part 1)

Each component has a specific role in making the Arduino UNO Countdown Timer work accurately and efficiently.



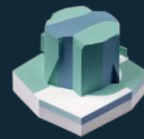
Arduino UNO

The brain of the project. It processes the program and controls all connected components.



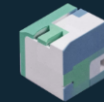
16×2 I2C LCD Display

Displays the countdown time clearly. The I2C module reduces wiring and makes the circuit simple.



Rotary Encoder

Used to set the countdown time by rotating the knob and pressing it to confirm the selection.

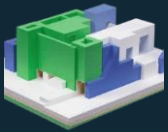


Active Buzzer

Gives a beep sound when the timer reaches zero to alert the user.

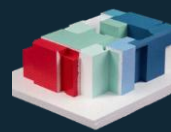
COMPONENTS USED

Hardware Components Used (Part 2)



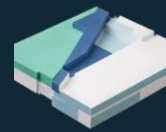
Green LED

Glows while the timer is running.



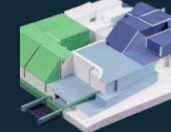
Red LED

Lights up when the timer reaches zero.



220Ω Resistor

Controls current and protects the LEDs.



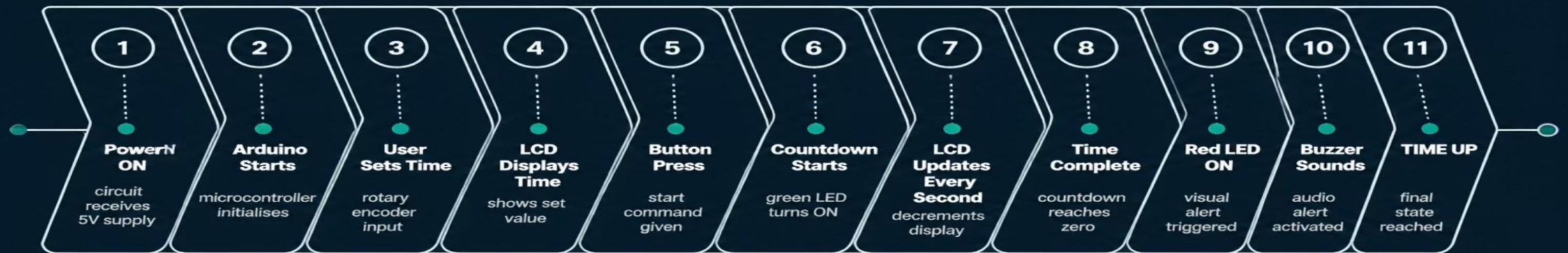
Jumper Wires

Connect all the components together.

Together, these parts make the Arduino UNO Countdown Timer accurate, reliable, and easy to use.

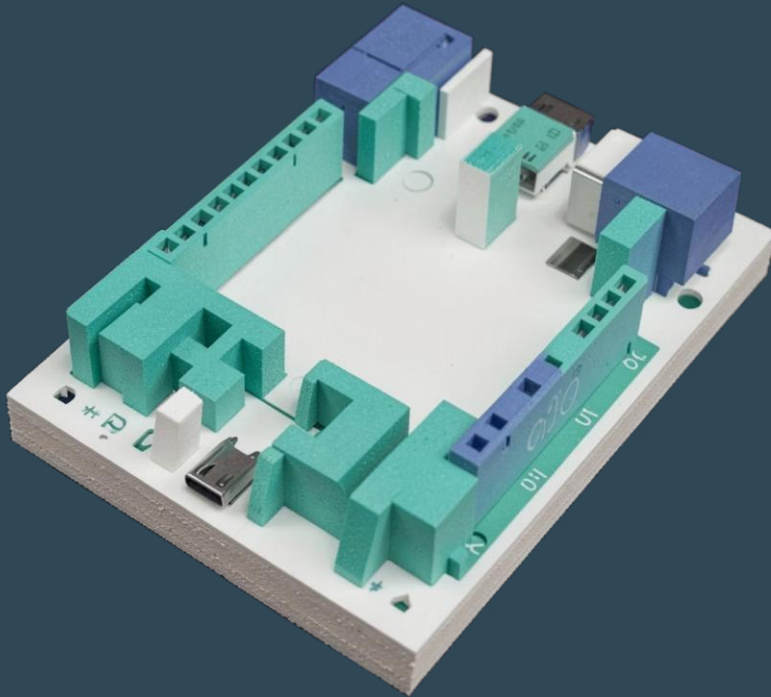
How the Timer Works — Step by Step

The timer follows a simple sequence, from setting the time to triggering an alert when the countdown ends.



COMPONENT DEEP DIVE

Understanding Each Component



Arduino UNO

What: Main controller board.

Why: Easy to program.

How: Runs the countdown.



LCD 16×2 I2C

What: Display screen.

Why: Shows time clearly.

How: Receives data by I2C.



Rotary Encoder

What: Input knob.

Why: Durable and easy to use.

How: Rotate to set time.



Green LED

What: Green indicator light.

Why: Shows timer is running.

How: Glows during countdown.



Red LED

What: Red indicator light.

Why: Shows timer has finished.

How: Lights up at time-up.



Buzzer

What: Sound alert device.

Why: Alerts user when time ends.

How: Beeps when triggered.

ADVANTAGES

Why This Timer Stands Out

Arduino UNO countdown timer for learning and everyday use.



Easy to Use

Set the time with one knob.
Press to start.



Portable

Compact design.
Works with USB or 9V battery.



Accurate

Counts down second by second.



Educational Project

Teaches Arduino, electronics, and
timers.



Reliable

No crashes, updates, or internet
needed.



User Friendly

Simple for students, teachers, and
seniors.

Where Can This Timer Be Used?

This countdown timer can be used in many real-life situations.



Kitchen Timer

Helps track
cooking time
accurately.



School & Study Sessions

Useful for study
sessions and
classroom
activities.



Sports & Workout

Perfect for
workout intervals
and sports drills.



Quiz Competitions

Sets a fixed time
limit for each
question.

FUTURE SCOPE

The Road Ahead — What's Next?

This project can be upgraded with additional modules in the future.



Bluetooth

Add an HC-05 module to control the timer wirelessly.



WiFi + IoT

Connect the timer to WiFi using an ESP8266 module.



Relay Automation

Control external devices automatically using a relay.



Battery Backup

Add a rechargeable battery for portable use.

Thank You

Thank you for your time and attention.
We hope this presentation gave you a clear understanding of our Arduino UNO Countdown Timer project.

Built With

Arduino UNO · LCD · Rotary Encoder
Module · LEDs · Buzzer

Programmed In

Arduino IDE (C++) — Open Source &
Free

Key Takeaway

Simple components + smart programming = useful real-world solutions.

Questions & Answers

Thank you for your time and attention!
We would be happy to answer your questions.

